Problem Statement No.15: The Daily Air Quality Action Agent

The Challenge

Air pollution is a **major public health crisis**, contributing to respiratory issues, climate change, and reduced quality of life. Citizens often receive air quality index (AQI) data, but it is presented in **technical formats**, making it difficult to understand or translate into **daily lifestyle actions**. Traditional monitoring systems lack **personalized recommendations** and proactive alerts.

The Objective

Build an **AI-powered Daily Air Quality Action Agent** that translates **air quality data into actionable insights** for citizens, policymakers, and organizations. The solution should:

1. **AQI Data Retrieval (RAG) –** Collect and interpret real-time AQI from public APIs, satellite data, or government sources.
2. **Personalized Health Recommendations** – Provide daily actionable advice (e.g., avoid outdoor exercise, use masks, keep children indoors) tailored to age, location, and health conditions.
3. **Predictive Alerts –** Forecast poor air quality days, issue early warnings, and suggest preventive measures like adjusting commute or using air purifiers.
4. **Policy & Awareness Insights –** Suggest community-level actions (e.g., carpooling, waste reduction) and summarize policy advisories for public awareness.
5. **Multi-Agent System** -
    - AQI Data Agent (fetches & interprets data)
    - Health Advisory Agent (personalized lifestyle suggestions)
    - Forecasting Agent (predicts trends & risks)
    - Community Action Agent (drives awareness campaigns)
6. **Visualization Dashboard** – Show daily AQI levels, risk alerts, and recommended actions in a clear and interactive format.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models